ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

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C01-2

EXPERIMENT 6

Aim:

To write program for capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constants using MATLAB.

Test case(

Determination and plotting of capacitance of parallel plate capacitor for given area and dielectric thickness

Program:

```
clc;
clear;
A=input("Enter the area value=");
 d=input("Enter the thickness value=");
epsilon0=8.854*10^-12; C=epsilon0*A/d;
 disp('The capacitance is');
 disp(C);
% Plot: capacitance versus plate separation for the given area dv = linspace(1e-3,10e-3,50);
Cv = epsilon0*A./dv;
figure; plot(dv*1e3,Cv,'LineWidth',2);
 grid on; xlabel('Plate separation d (mm)');
 ylabel('Capacitance (F)');
title('Parallel plate capacitance versus separation');
 % Validation using Laplace's equation (1-D finite-difference solution between the plates) V0 = 100;
N = 51;
V = zeros(1,N);
V(N) = V0;
for it = 1:20000 V(2:N-1) = 0.5*(V(1:N-2) + V(3:N));
% Laplace equation: d2V/dx2 = 0 end x = linspace(0,d,N);
E = -gradient(V,x);
```

% E = -dV/dx C_lap = epsilon0*A*abs(mean(E))/V0;

% C = Q/V0 with Q = epsilon0*E*A fprintf('Capacitance from Laplace solution = %.4e F\n', C_lap);

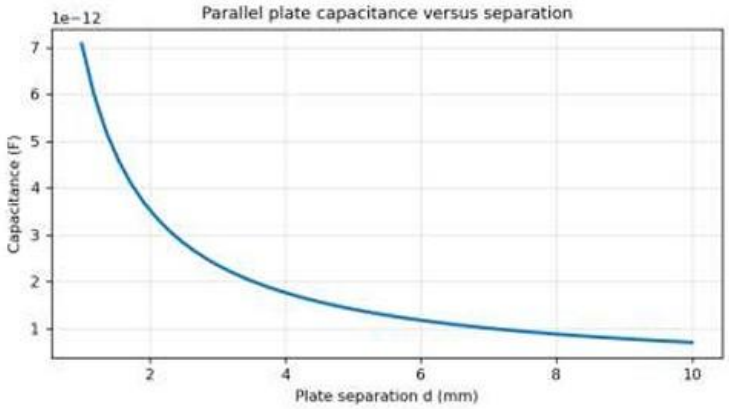
Output:

Enter the area value = 8e-4

Enter the thickness value = 3e-3

The capacitance is 2.36 × 10^{-12} F

Capacitance from Laplace solution = 2.3611e-12 F



Test case

(ii) Determination of capacitance of parallel plate capacitor with three different dielectric layers

Program:

clc;

A=input("Enter the area value=");

d1=input("Enter thickness d1=");

d2=input("Enter thickness d2=");

d3=input("Enter thickness d3=");

er1=input("Enter dielectric constant er1=");

er2=input("Enter dielectric constant er2=");

er3=input("Enter dielectric constant er3=");

epsilon0=8.854e-12;

C=epsilon0*A/((d1/er1)+(d2/er2)+(d3/er3));

disp('The capacitance is');

```
disp(C);
```

OUTPUT:

Enter the area value = 8e-4

Enter thickness d1 = 1e-3

Enter thickness d2 = 3e-3

Enter thickness d3 = 2e-3

Enter dielectric constant er1 = 2.3

Enter dielectric constant er2 = 3.3

Enter dielectric constant er3 = 4

The capacitance is 3.84×10^{-12} F

```
Enter the area value=8e-4
Enter thickness d1=1e-3
Enter thickness d2=3e-3
Enter thickness d3=2e-3
Enter dielectric constant er1=2.3
Enter dielectric constant er2=3.3
Enter dielectric constant er3=4
The capacitance is
   3.8415e-12
```

Test case (iii)

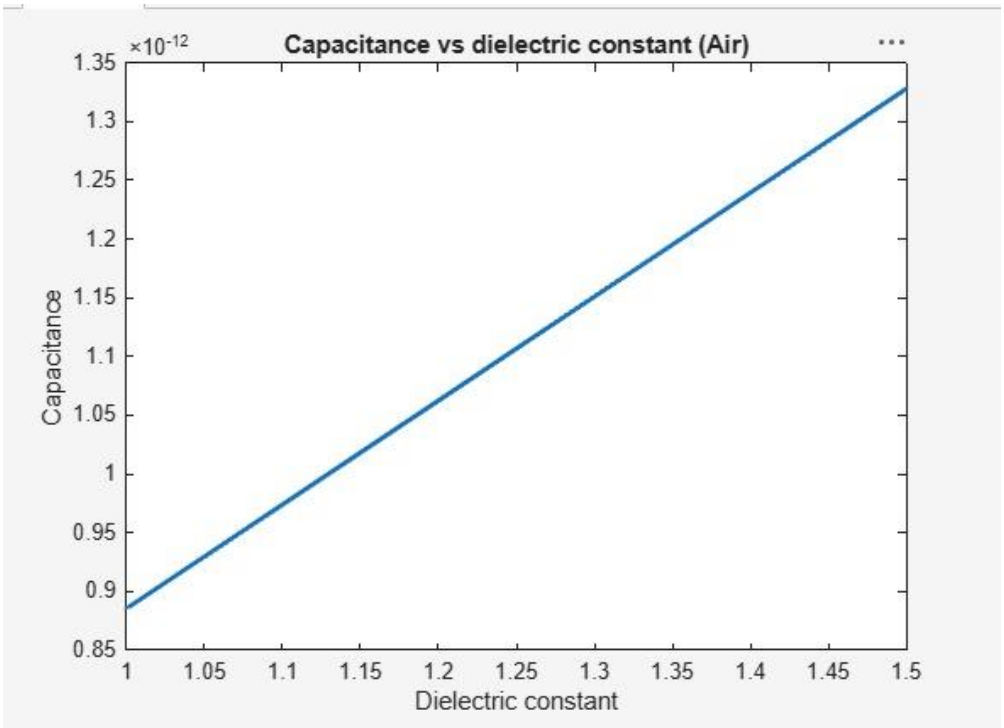
Determination and plotting of capacitance variation with dielectric constant (Air)

Program:

```
clc;
epsilon0 = 8.854e-12;
A = 1e-4; d = 1e-3;
er = linspace(1,1.5,100);
C = (epsilon0 .* er .* A) ./ d;
plot(er,C,'LineWidth',2);
xlabel('Dielectric constant');
ylabel('Capacitance');
title('Capacitance vs dielectric constant (Air)');
```

Output:

Capacitance increases linearly with dielectric constant



Test case (iv)

Determination and plotting of capacitance variation with dielectric constant (Glass)

```
clc;

epsilon0 = 8.854e-12;

A = 1e-4; d = 1e-3;

er = linspace(4,8,100);

C = (epsilon0 .* er .* A) ./ d;

plot(er,C,'LineWidth',2);

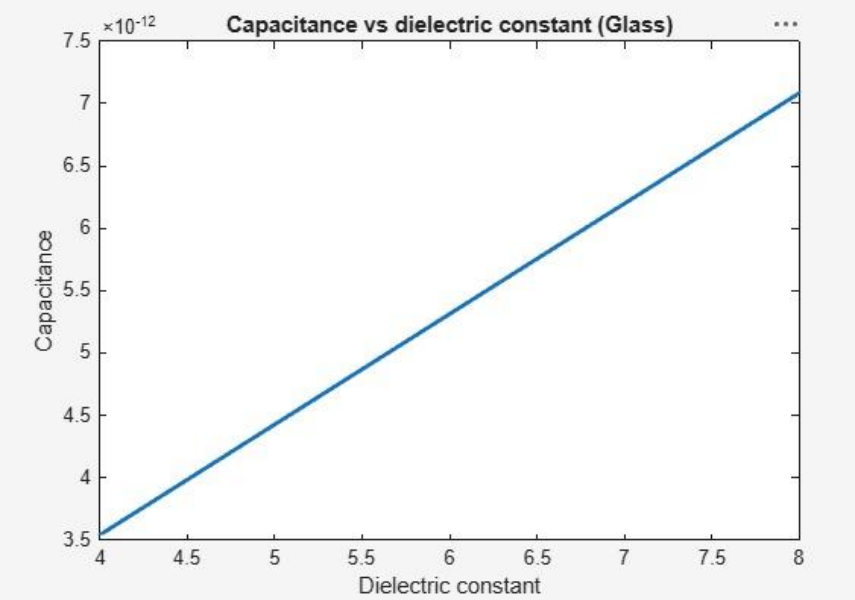
xlabel('Dielectric constant');

ylabel('Capacitance');

title('Capacitance vs dielectric constant (Glass)');
```

Output:

Higher dielectric → higher capacitance



RESULTS:

Thus the program was verified for capacitance of parallel plate capacitor in different dielectric medium using MATLAB successfully.

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Set capacitor geometry and material properties	Set capacitor geometry and material properties	Correctly enters plate area, separation and relative permittivity with SI units.	Inputs are correct with a minor unit/ property issue.	Several inputs need correction.	Geometry or dielectric inputs are invalid.	25	2.5	
2	Calculate capacitance in MATLAB	Calculate capacitance in MATLAB	Uses the correct capacitance relation and calculates values for all specified media.	Calculations are mostly correct with minor errors.	Only some cases are calculated correctly.	Capacitance calculations are incorrect.	25	2.5	
3	Compare dielectric and geometry effects	Compare dielectric and geometry effects	Correctly evaluates trends versus permittivity/area/ separation and validates the model.	Trend and validation are mostly correct with minor omissions.	Trend discussion or validation is incomplete.	Incorrectly interprets parameter effects.	30	3	
4	Document engineering observations	Document engineering observations	Provides a complete table/plot with units and	Documentation and explanation	Documentation and explanation are	Results are incomplete or unexplained.	20	2	

			explains dielectric influence.	are mostly complete.	mostly complete.				
Total							100	10	